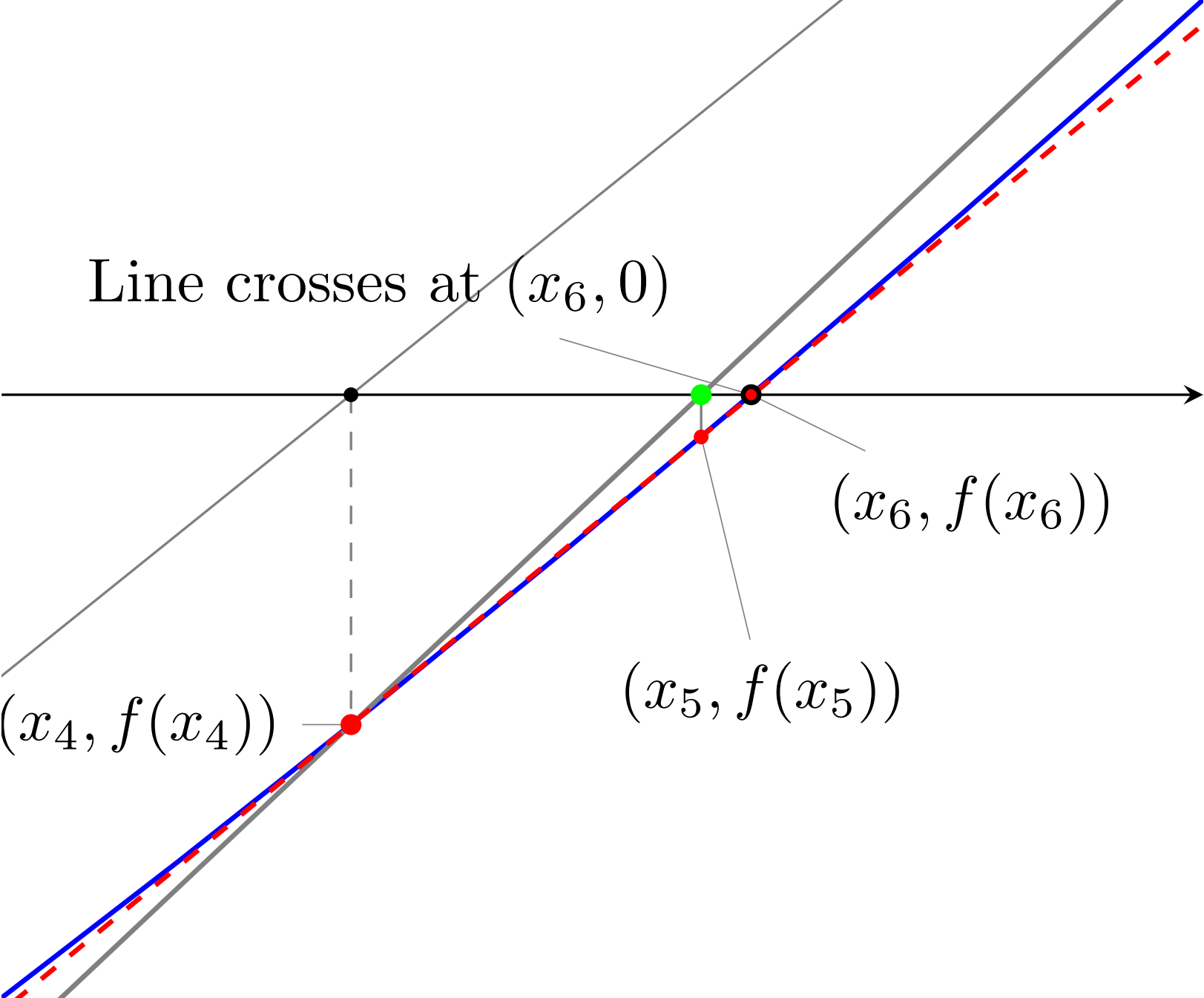
Line crosses at $(x_6, 0)$
$(x_6, f(x_6))$
$(x_5, f(x_5))$
$(x_4, f(x_4))$